

Calculator Model / No.

Name: _____

Class: _____

Reg Number: _____



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Chemistry

9647/02

Paper 2 Structured Questions

18 September 2015**2 hours**Additional Materials: *Data Booklet*

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.
Write your calculator brand and model/number in the box provided above.

Answer **all** questions in the spaces provided on the question paper.
All working must be shown clearly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE		
Paper 1	MCQ	/ 40
Paper 2	Q1	/ 12
	Q2	/ 13
	Q3	/ 10
	Q4	/ 10
	Q5	/ 13
	Q6	/ 14
		/ 72
Paper 3		/ 80
Total		/ 192
Grade		

Answer **all** questions in the space provided.

1 Planning (P)

Ethanol is widely used in industries as a solvent and as a precursor for other organic substances such as ethanal. Similarly, ethanal can also be used as a substance to synthesise other organic compounds including ethanoic acid.

The boiling points of the three compounds are as follows:

compound	ethanol	ethanal	ethanoic acid
boiling points / °C	78	20	118

A chemist carried out an experiment to synthesise ethanal from ethanol using the procedure described below. The reaction was found to be highly exothermic.

About 5.0 g of potassium dichromate (an excess) is dissolved in 6 cm³ of water together with 5 cm³ of ethanol. This is then added dropwise to 9 cm³ of dilute sulfuric acid solution, which has been pre-heated to about 60 °C, in a round-bottom flask. The mixture is not heated during the reaction and the product is collected immediately using an ice-bath.

(a) Suggest why heating of the reaction mixture is **not** required during the reaction.

[1]

(b) (i) Write an equation for the oxidation of ethanol to form ethanal. Use [O] to represent the oxidising agent.

[1]

(ii) The density of ethanol is 0.789 g cm⁻³. Calculate the theoretical mass of ethanal that can be produced from the above experiment.

[1]

[Turn over

- (c)** Using the information provided as well as your answer in **(b)**, write a plan to synthesise ethanal and determine its percentage yield by mass.

In your plan, you need to:

- draw a labelled diagram of the assembled apparatus you would use for synthesis of ethanal;
- give a full description of the procedures you would use to prepare ethanal;
- describe how the product is collected during the reaction;
- describe how to ascertain that the reaction has reached completion;
- explain how you would determine the percentage yield of ethanal by mass.

- (d) The ethanal collected may be contaminated with ethanoic acid and sulfuric acid. Suggest a reagent that could be used to remove these impurities.

[1]

- (e) Identify **one** potential safety hazard in this experiment and state how you would minimise this risk.

[1]

[Total: 12]

- 2 The elements across Period 3 show a change from metallic to non-metallic nature, which is reflected in the bonding of their compounds.

- (a) (i) Explain the variation in atomic radius across Period 3.

[2]

- (ii) Period 3 elements react with oxygen to form the corresponding oxides of varying acid-base nature.

Write an equation for the reaction of each of the following oxides with aqueous sodium hydroxide.

Sulfur dioxide:

Aluminium oxide:

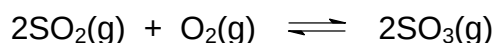
[2]

[Turn over]

(iii) Explain the acid-base nature of sulfur dioxide and aluminum oxide.

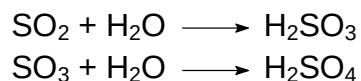
[2]

(b) The key stage in the manufacture of sulfuric acid is the reaction between sulfur dioxide and oxygen.



In an experiment, a mixture of 0.24 moles of SO_2 and 0.12 moles of O_2 was allowed to reach equilibrium at $T^\circ\text{C}$ and a total pressure of 6 atm. The equilibrium number of moles of SO_3 formed is found to be x moles.

The gases were then rapidly cooled and an excess of water was added to the mixture of gases, under suitable conditions, causing the following reactions to occur.



You may assume that H_2SO_3 behaves as a strong monobasic acid.

(i) Determine, in terms of x , the number of moles of H^+ that will be formed after the addition of water to the mixture of SO_2 and SO_3 .

[2]

[Turn over]

The resulting solution was made up to 250 cm³ in a standard flask. 25.0 cm³ of this solution was titrated with 1.00 mol dm⁻³ NaOH solution and required 38.50 cm³ for neutralisation.

- (ii) Calculate the number of moles of H⁺ ions present in the 250 cm³ standard flask.

[1]

- (iii) Use your answers in **(b)(i)** and **(b)(ii)** to calculate the amounts, in moles, of SO₂, O₂ and SO₃ in equilibrium at T °C for the reversible reaction between sulfur dioxide and oxygen.

Hence, calculate a value for K_p for the above reversible reaction.

[4]

[Total: 13]

[Turn over

- 3 (a)** In an experiment, a sample of chloride ions was vapourised and passed through an electric field. Analysis revealed that a beam of $^{37}\text{Cl}^-$ gives an angle of deflection of -1.0° .

- (i)** In the same experimental set-up, by what angle will a beam of helium nuclei be deflected?

[1]

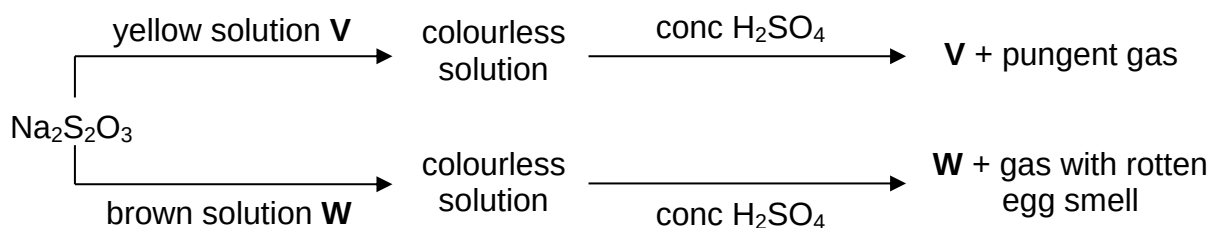
- (ii)** Under identical conditions, a beam of particles, **Z**, each having 32 times the mass of a proton, was deflected by an angle of -2.3° . Deduce the identity of **Z**.

[2]

- (b)** State and explain the trend of first ionisation energy of halogens down Group VII.

[2]

- (c) Although halogens and their compounds can be toxic, a number of them are essential for the functioning of the human body and are utilised in everyday products. The reaction scheme below shows the reactions involving some of the halogens and their compounds.



- (i) Suggest the identities of **V** and **W**.

[1]

- (ii) Write a balanced equation for the reaction between $\text{Na}_2\text{S}_2\text{O}_3$ and **W**.

[1]

- (iii) State the role of $\text{Na}_2\text{S}_2\text{O}_3$ in the reaction in (c)(ii).

[1]

- (d) When halogens are reacted with hydrogen gas, hydrogen halides are produced. Data about hydrogen halides are given below.

	HF	HCl	HBr	HI
boiling point / °C	+20	-85	-67	-35
K_a / mol dm ⁻³	5.6×10^{-4}	1.0×10^{-7}	1.0×10^{-9}	1.0×10^{-11}

Explain why the K_a of hydrogen halides increases down the group.

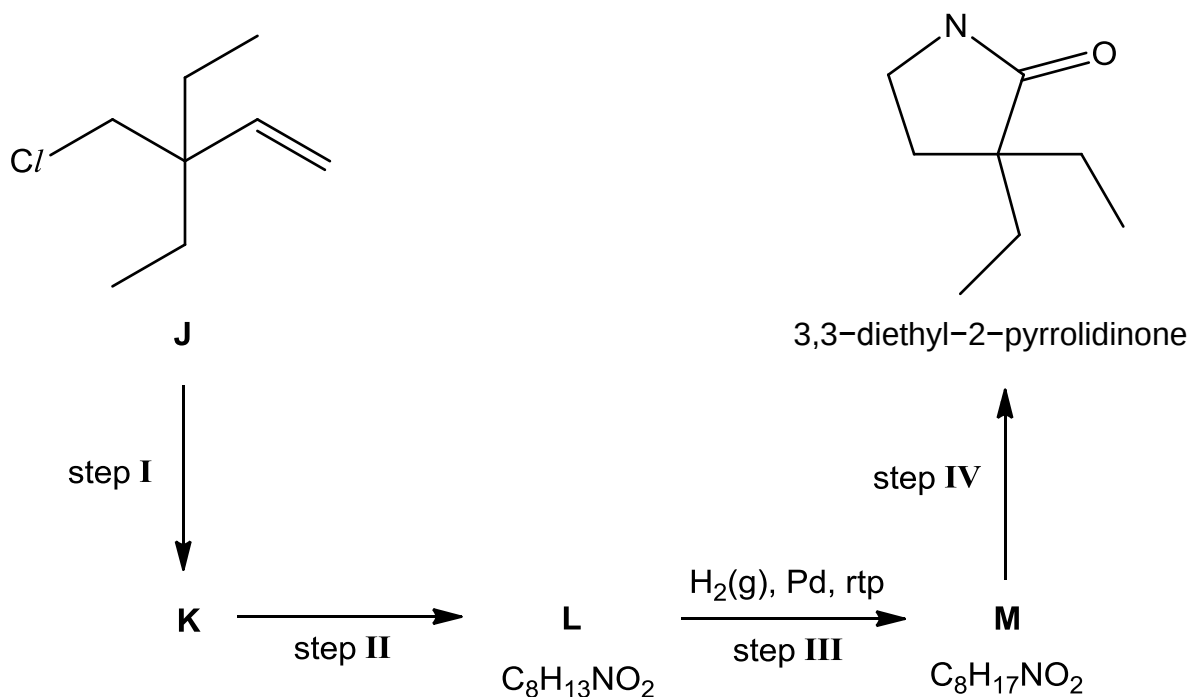
[2]

[Total: 10]

[Turn over]

- 4 (a) 3,3-diethyl-2-pyrrolidinone is a anticonvulsant drug that extends lifespan in nematode worms. A scientist attempted to synthesise 3,3-diethyl-2-pyrrolidinone starting from compound **J**.

The suggested synthetic scheme is as shown below.



- (i) Suggest the reagents and conditions required for the following steps:

Step I:

Step II:

Step IV:

[3]

- (ii) Suggest the structures of **K** and **M**.



[2]

- (b)** Most silver salts such as silver chromate(VI), Ag_2CrO_4 are sparingly soluble in water.

In an experiment, 8.12×10^{-3} mol of potassium chromate(VI), K_2CrO_4 , was dissolved in 500 cm^3 of a saturated solution of Ag_2CrO_4 . The solubility product of Ag_2CrO_4 is $3.01 \times 10^{-12} \text{ mol}^3 \text{ dm}^{-9}$ at 25°C .

- (i)** Calculate the solubility of Ag_2CrO_4 in water **before** the addition of K_2CrO_4 .

[1]

- (ii)** Calculate the solubility of Ag_2CrO_4 **after** the dissolution of K_2CrO_4 .

[2]

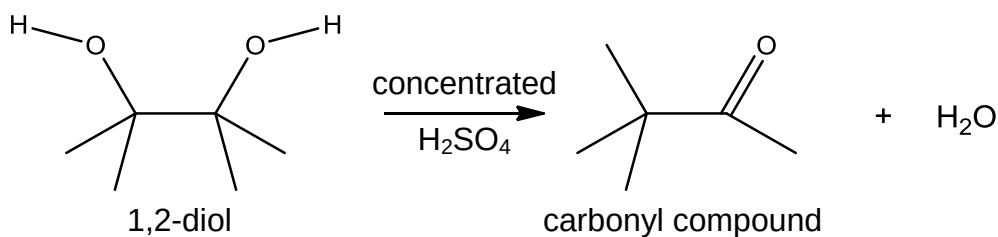
- (iii)** Use your answers in part **(b)(i)** and **(b)(ii)** to calculate the mass of Ag_2CrO_4 precipitated after the dissolution of K_2CrO_4 .

[2]

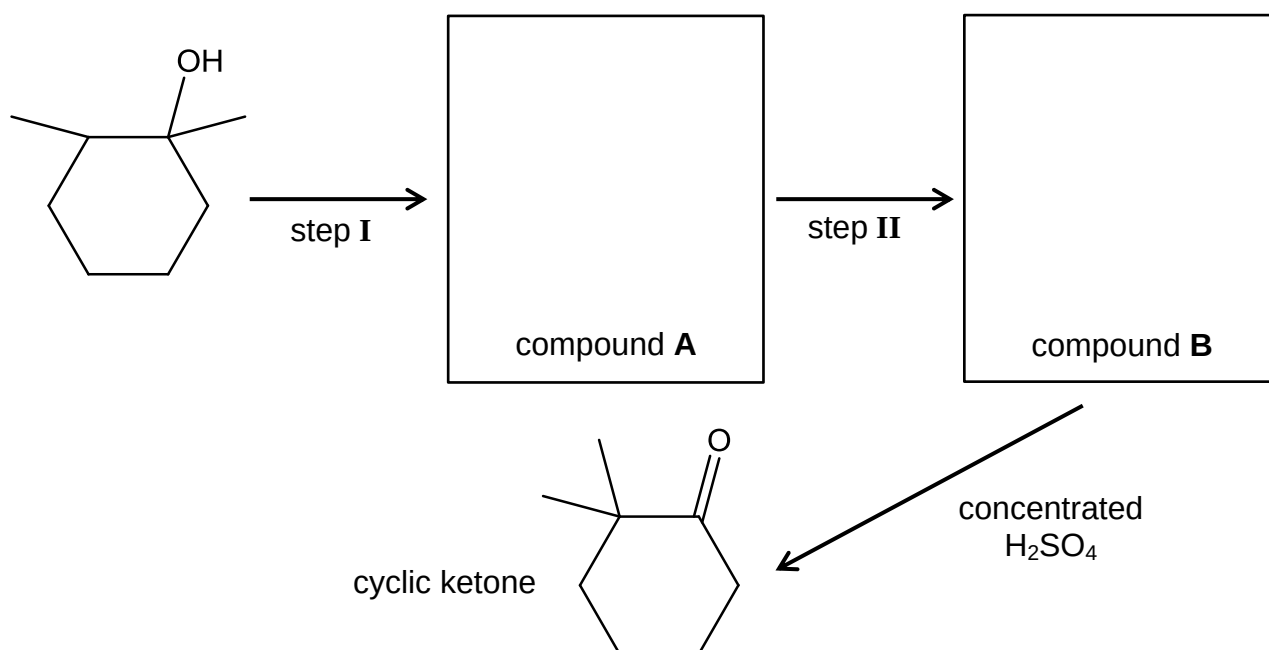
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- 5 The pinacol rearrangement reaction, first described by Wilhelm Rudolph Fittig in 1860 involves the conversion of a 1,2-diol to a carbonyl compound through the use of concentrated sulfuric acid. The mechanism for this reaction is as such:



The following illustrates the synthetic route for the formation of a cyclic ketone.



- (a) (i) Draw the structures of compounds **A** and **B** in the boxes provided.

[2]

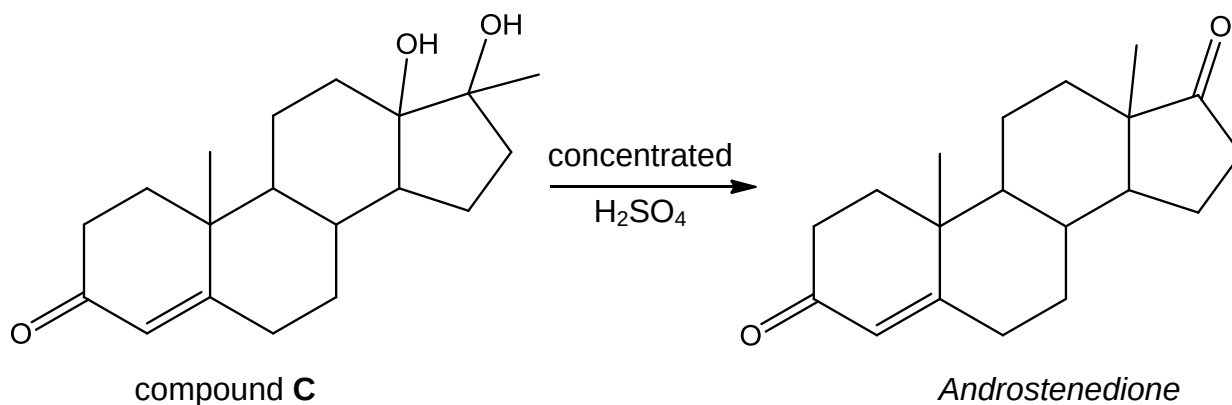
- (ii) State the reagents and conditions required for steps I and II.

Step I :

Step II :

[2]

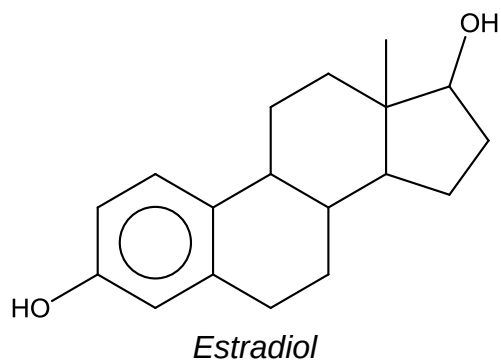
- (b) Alcohol functional groups are found in numerous hormones, such as *Androstenedione*. Synthetically, *Androstenedione* may be synthesised from the pinacol rearrangement reaction using compound **C** as a starting material.



Suggest one chemical test that would enable compound **C** and *Androstenedione* to be distinguished from each other.

[2]

- (c) Estradiol is an estrogenic hormone.



Draw the structure of the organic product formed when *Estradiol* is reacted with each of the following under suitable conditions:

- (i) $\text{Cl}_2(\text{aq})$

- (ii) PCl_5

- (iii) hot alkaline KMnO_4

- (d) Benzene may be used as a starting material for the synthesis of *Estradiol*.

A scientist carried out an experiment to generate a dimethylbenzene from benzene. When the dimethylbenzene is nitrated by a mixture of concentrated nitric acid and concentrated sulfuric acid, the products contained a number of isomers of molecular formula $C_8H_9NO_2$.

One of the isomers, **P**, has a plane of symmetry perpendicular to the plane of the benzene ring in its molecule. The formation of **P** proceeds via an organic intermediate **R**.

- (i) Draw the displayed formulae of **P** and **R**, showing clearly the geometry of the bonds around the carbon atom bonded to the nitrogen atom.

Isomer P	Intermediate R

[2]

- (ii) State and outline the mechanism for the formation of **P** from the dimethylbenzene.

Name of mechanism :

[2]

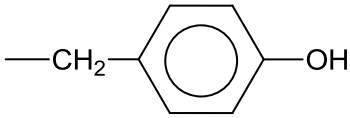
[Total : 13]

[Turn over]

- 6 Proteins can be classified as primary, secondary, tertiary and quaternary structures. The tertiary structure of proteins enables it to adopt a 3-dimensional shape which allows it to be soluble in water and be mobile within the cell.

Table 1 below shows the R groups of various α -amino acids present in polypeptide chains.

Table 1

α -amino acid	M_r	formula of side-chain (R in $RCH(NH_2)COOH$)
asp	133	$-CH_2CO_2H$
lys	146	$-CH_2CH_2CH_2CH_2NH_2$
leu	131	$-CH_2CH(CH_3)_2$
cys	121.1	$-CH_2SH$
tyr	181	
gln	146	$-CH_2CH_2CONH_2$

- (a) Explain what is meant by an α -amino acid.

[1]

- (b) Globular proteins are more water soluble than other classes of proteins.

- (i) Some of the amino acid residues shown in **Table 1** would enable globular proteins to be soluble in water. Identify suitable amino acids residues to illustrate **two** different types of interactions between the selected amino acids and water molecules.

[2]

[Turn over

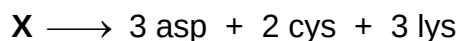
- (ii) Explain how the R groups of the amino acid residues listed in **Table 1** are arranged in a globular protein.

[2]

- (iii) The secondary structure of globular proteins consist of segments of α -helices. Describe, with the aid of a diagram, how the polypeptide chain is held in the structure of an α -helix.

[3]

- (c) When a polypeptide **X** undergoes hydrolysis, three aspartic acids, two cysteine and three lysine molecules were formed as shown in the following reaction:



Calculate the M_r of polypeptide **X**.

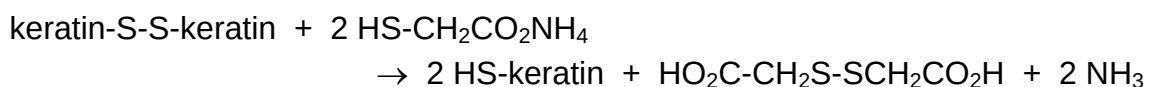
[1]

- (d) Hair comprises primarily of the protein keratin. Curly hair is due to covalent disulfide bonds which can be formed between the amino acid residues by an oxidation process.

- (i) Write a chemical equation to show how the cysteine residues in the keratin molecule can form disulfide bridges.

[1]

- (ii) In order to straighten curly hair permanently, using hair rebonding method is more viable. Rebonding hair is a chemical treatment with ammonium thioglycolate, $\text{HSCH}_2\text{CO}_2\text{NH}_4$ used to straighten hair. It can reverse the reaction in (d)(i) and break the disulfide bridges in hair as shown.



What aspect of keratin's protein structure is being altered during the process of rebonding?

[1]

- (iii) Different hair-straightening methods target various chemical bonds that hold keratin strands together. Keratin forms a long helical chain like a very thin strand of rope.

Based on the helical structure of keratin, suggest why using a heating iron may not straighten curly hair permanently.

[1]

- (iv) Upon hydrolysis of keratin, a tripeptide, *gln-tyr-lys* is obtained. Draw the structural formula of this tripeptide at pH 1.

[2]

[Total : 14]

END OF PAPER